

0.1 Riemann-Stieltjes Integral

Definition 0.1.1. A finite subset $P \subset [a, b]$ containing a, b is called a *Partition* of $[a, b]$. Conventionally, denote $P = \{x_0, x_1, \dots, x_n\}$ where $a = x_0 < x_1 < \dots < x_n = b$. The set of Partitions of $[a, b]$ is denoted as $\mathcal{P}[a, b]$.

Definition 0.1.2. Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha: [a, b] \rightarrow \mathbb{R}$ be a monotone-increasing. Let a partition $P \in \mathcal{P}[a, b]$ be given, and denote $P = \{x_0, x_1, \dots, x_n\}$. For each $1 \leq i \leq n$, define

$$\Delta\alpha_i \stackrel{\text{def}}{=} \alpha(x_i) - \alpha(x_{i-1}) \underset{\alpha \text{ mono.}}{\geq} 0$$

and

$$M_i \stackrel{\text{def}}{=} \sup\{f(t) \mid x_{i-1} \leq t \leq x_i\} \text{ and } m_i \stackrel{\text{def}}{=} \inf\{f(t) \mid x_{i-1} \leq t \leq x_i\}$$

And, define *Upper Sum* and *Lower Sum* as:

$$U(f, P, \alpha) \stackrel{\text{def}}{=} \sum_{i=1}^n M_i \cdot \Delta\alpha_i \text{ and } L(f, P, \alpha) \stackrel{\text{def}}{=} \sum_{i=1}^n m_i \cdot \Delta\alpha_i$$

Now, define *Upper integral* and *Lower integral* as:

$$\overline{\int_a^b} f d\alpha \stackrel{\text{def}}{=} \inf_{P \in \mathcal{P}[a, b]} U(f, P, \alpha) \text{ and } \underline{\int_a^b} f d\alpha \stackrel{\text{def}}{=} \sup_{P \in \mathcal{P}[a, b]} L(f, P, \alpha)$$

We say that f is *Riemann-Stieltjes Integrable* with respect to α if:

$$\overline{\int_a^b} f d\alpha = \underline{\int_a^b} f d\alpha$$

The set of Riemann-Stieltjes Integrable functions with respect to α as $\mathcal{R}(\alpha)$.

Lemma 0.1.3. Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha: [a, b] \rightarrow \mathbb{R}$ be a monotone-increasing. For any partitions $P, Q \subset [a, b]$, and a refinement $P^* \supset P$,

$$L(f, P, \alpha) \leq L(f, P^*, \alpha) \leq \underline{\int_a^b} f d\alpha \leq \overline{\int_a^b} f d\alpha \leq U(f, P^*, \alpha) \leq U(f, P, \alpha)$$

and

$$L(f, P, \alpha) \leq U(f, Q, \alpha)$$

Proof of Lemma 0.1.3.

Let $P^* \supset P$ be a refinement of P . Since partition contains only finite elements, WLOG, assume that $P^* = P \cup \{s\}$ s.t. $s \notin P = \{x_0, \dots, x_n\}$

Then, for some $1 \leq j \leq n$ $x_{j-1} < s < x_j$. Now,

$$\begin{aligned} U(f, P, \alpha) - U(f, P^*, \alpha) &= \sup_{t \in [x_{j-1}, x_j]} f(t) \cdot [\alpha(x_j) - \alpha(s)] + \sup_{t \in [x_{j-1}, x_j]} f(t) \cdot [\alpha(s) - \alpha(x_{j-1})] \\ &\quad - \sup_{t \in [x_{j-1}, s]} f(t) \cdot [\alpha(x_j) - \alpha(s)] - \sup_{t \in [s, x_j]} f(t) \cdot [\alpha(s) - \alpha(x_{j-1})] \\ &\geq 0. \end{aligned}$$

0.1.1 Properties of Riemann-Stieltjes Integral

Theorem 0.1.4. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \rightarrow \mathbb{R}$ be monotone-increasing. TFAE:

1. f is Riemann-Stieltjes Integrable with respect to α on $[a, b]$.
2. For any $\varepsilon > 0$, there exists a Partition $P \in \mathcal{P}[a, b]$ such that $U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon$.
3. For some $A \in \mathbb{R}$, for any $\varepsilon > 0$, there exists $P_0 \in \mathcal{P}[a, b]$ such that

$$\forall P \in \mathcal{P}[a, b], P \supseteq P_0 \implies |S(f, P, \alpha) - A| < \varepsilon$$

where Riemann-Stieltjes Sum $S(f, P, \alpha) \stackrel{\text{def}}{=} \sum_{i=1}^n f(t_i)[\alpha(x_i) - \alpha(x_{i-1})]$, for some $t_i \in [x_{i-1}, x_i]$.

Lemma 0.1.5. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \rightarrow \mathbb{R}$ be a monotone-increasing.

If $f, g \in \mathcal{R}(\alpha)$ and $c \in \mathbb{R}$, then $f + g, cf \in \mathcal{R}(\alpha)$ with

$$\begin{aligned}\int_a^b (f + g) d\alpha &= \int_a^b f d\alpha + \int_a^b g d\alpha \\ \int_a^b (cf) d\alpha &= c \int_a^b f d\alpha\end{aligned}$$

Lemma 0.1.6. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \rightarrow \mathbb{R}$ be a monotone-increasing, and $a < c < b$.

f is Stieltjes Integrable on $[a, b]$ if and only if f is Stieltjes Integrable on $[a, c]$ and $[c, b]$

In this case, the following equality holds:

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha$$

Lemma 0.1.7. Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $\alpha, \beta : [a, b] \rightarrow \mathbb{R}$ be monotone-increasing.

If $f \in \mathcal{R}(\alpha)$, $f \in \mathcal{R}(\beta)$ and $c > 0$, then $f \in \mathcal{R}(\alpha + \beta)$ and $f \in \mathcal{R}(c\alpha)$ and $|f| \in \mathcal{R}(\alpha)$ with

$$\begin{aligned}\int_a^b f d(\alpha + \beta) &= \int_a^b f d\alpha + \int_a^b f d\beta \\ \int_a^b f d(c\alpha) &= c \int_a^b f d\alpha \\ \left| \int_a^b f d\alpha \right| &\leq \int_a^b |f| d\alpha\end{aligned}$$

Proof of Thm 0.1.4

1) $\Rightarrow$ 2). Since $f \in \mathcal{R}(a)$, given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a, b]$ s.t.

$$\int_a^b f \, d\alpha \leq U(f, P, \alpha) < \int_a^b f \, d\alpha + \frac{\varepsilon}{2} \quad \text{and} \quad \int_a^b f \, d\alpha \geq L(f, P, \alpha) > \int_a^b f \, d\alpha - \frac{\varepsilon}{2}$$

Ask the next step to a toddler on the street.

2) $\Rightarrow$ 1). It is clear, because for any partition $P \in \mathcal{P}[a, b]$,

$$\int_a^b f \, d\alpha - \int_a^b f \, d\alpha \leq U(f, P, \alpha) - L(f, P, \alpha)$$

To show 1) $\Leftrightarrow$ 3), we prove a statement: If 2) holds, then for any $1 \leq i \leq n$,

$$\text{for arbitrary } s_i, t_i \in [x_{i-1}, x_i], \quad \sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i < \varepsilon.$$

$$\text{Since } |f(s_i) - f(t_i)| \leq M_i - m_i,$$

$$\sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i \leq U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

1) $\Rightarrow$ 3). Suppose $f \in \mathcal{R}(a)$. Then, by 2), there exists a partition $P_0 \in \mathcal{P}[a, b]$ s.t.

$$U(f, P_0, \alpha) - L(f, P_0, \alpha) < \varepsilon$$

Now, for any refinement $P > P_0$,

$$U(f, P, \alpha) - L(f, P, \alpha) \leq U(f, P_0, \alpha) - L(f, P_0, \alpha) < \varepsilon$$

Clearly, regardless of choice of t_i ,

$$L(f, P, \alpha) \leq S(f, P, \alpha) \leq U(f, P, \alpha)$$

$$\text{and} \quad U(f, P, \alpha) \geq \int f \, d\alpha > L(f, P, \alpha)$$

$$\text{Therefore, } |S(f, P, \alpha) - \int f \, d\alpha| \leq U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

3) $\Rightarrow$ 1) is trivial.

Proof of Lemma 0.1.5.

Let $f, g \in \mathcal{R}(a)$ be given. Given $\epsilon > 0$, there exists a partition $P \in \mathcal{P}[a, b]$ such that

$$U(f, P, \alpha) - L(f, P, \alpha) < \frac{\epsilon}{2} \quad \text{and} \quad U(g, P, \alpha) - L(g, P, \alpha) < \frac{\epsilon}{2}$$

(Technically, take $P_1, P_2 \in \mathcal{P}[a, b]$ for f, g respectively, and put $P = P_1 \cup P_2$). Now,

$$U(f+g, P, \alpha) - L(f+g, P, \alpha) \leq U(f, P, \alpha) + U(g, P, \alpha) - L(f, P, \alpha) - L(g, P, \alpha) < \epsilon.$$

And the equality is obtained by: given $\epsilon > 0$, there exists a partition $P \in \mathcal{P}[a, b]$ s.t.

$$\begin{aligned} \int_a^b f \, d\alpha + \int_a^b g \, d\alpha - \epsilon &\leq L(f, P, \alpha) + L(g, P, \alpha) \leq L(f+g, P, \alpha) \\ &\leq \int_a^b f+g \, d\alpha \\ &\leq U(f+g, P, \alpha) \leq U(f, P, \alpha) + U(g, P, \alpha) \leq \int_a^b f \, d\alpha + \int_a^b g \, d\alpha + \epsilon \end{aligned}$$

Proof of Lemma 0.1.6.

Suppose that $f \in \mathcal{R}(a)$ on $[a, b]$. Given $\epsilon > 0$, there exists $P \in \mathcal{P}[a, b]$ s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \epsilon.$$

Put $P^* = P \cup \{c\}$. Then, for $P_1 = P^* \cap [a, c]$ and $P_2 = P^* \cap [c, b]$. Now,

$$\begin{aligned} \int_a^c f \, d\alpha - \int_a^c f \, d\alpha + \int_c^b f \, d\alpha - \int_c^b f \, d\alpha &\leq U(f, P_1, \alpha) - L(f, P_1, \alpha) + U(f, P_2, \alpha) - L(f, P_2, \alpha) \\ &= U(f, P^*, \alpha) - L(f, P^*, \alpha) < \epsilon. \end{aligned}$$

Similarly, the converse holds.

Thm. Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded map and continuous except finitely many points. Then f is Riemann integrable.

Proof. Define $P_n = [a, b] \cap \left\{ \frac{k}{2^n} \mid k \in \mathbb{Z} \right\}$. Let A be a set of discontinuous points of f .

Put N be a number of elements of A . Now, for any $n \in \mathbb{N}$, there is at most $2N$ for $i \in \mathbb{N}$

$$\text{s.t. } A \cap [x_{i-1}, x_i] \neq \emptyset$$

where $x_i \in P_n$. Given $\varepsilon > 0$, take $n \in \mathbb{N}$ s.t. $2 \cdot \sup |f| \cdot 2N \cdot \frac{1}{2^n} < \frac{\varepsilon}{2}$.

Meanwhile, f is continuous on the set

$$C = \bigcup_{A \cap [x_{i-1}, x_i] = \emptyset} [x_{i-1}, x_i]$$

Therefore there is a partition P' on C s.t. $U(f, P') - L(f, P') < \frac{\varepsilon}{2}$.

Finally, for $P = P_n \cup P'$,

$$U(f, P) - L(f, P) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} < \varepsilon. \quad \square$$

0.1.2 Riemann-Stieltjes Integrable

Theorem 0.1.8. If $f: [a, b] \rightarrow \mathbb{R}$ is continuous, then $f \in \mathcal{R}(\alpha)$ for any monotone increasing $\alpha: [a, b] \rightarrow \mathbb{R}$.

Theorem 0.1.9. If $f: [a, b] \rightarrow \mathbb{R}$ is monotone and $\alpha: [a, b] \rightarrow \mathbb{R}$ is monotone increasing continuous, then $f \in \mathcal{R}(\alpha)$.

Theorem 0.1.10. If $f: [a, b] \rightarrow \mathbb{R}$ is map with $m \leq f(t) \leq M$ for all $t \in [a, b]$ and ϕ is continuous on $[m, M]$, then $\phi \circ f$ is Riemann-Stieltjes integrable with respect to α on $[a, b]$

Theorem 0.1.11. If $\alpha: [a, b] \rightarrow \mathbb{R}$ is continuously differentiable monotone increasing, then

$$f \in \mathcal{R}(\alpha) \text{ if and only if } f\alpha' \in \mathcal{R}(x)$$

In this case,

$$\int_a^b f(x) d\alpha(x) = \int_a^b f(x) \alpha'(x) dx$$

Proof of Thm 0.1.8.

Given $\epsilon > 0$, take $\eta > 0$ s.t. $[\alpha(b) - \alpha(a)] \cdot \eta < \epsilon$.

Since f is continuous on the compact set $[a, b]$, it is continuous uniformly.

Thus, there exists $\delta > 0$ s.t.

$$\forall x, y \in [a, b], |x - y| < \delta \Rightarrow |f(x) - f(y)| < \eta.$$

Now, let $P \in \mathcal{P}[a, b]$ be a partition such that for all i , $\Delta x_i < \delta$ ($P = \{x_0, \dots, x_n\}$)

Then, for all i , $M_i - m_i \leq \eta$. Now,

$$U(f, P, \alpha) - L(f, P, \alpha) = \sum_{i=1}^n (M_i - m_i) \cdot \Delta \alpha_i \leq \sum_{i=1}^n \eta \cdot \Delta \alpha_i = \eta \cdot \sum_{i=1}^n \Delta \alpha_i = \eta \cdot [\alpha(b) - \alpha(a)] < \epsilon.$$

Proof of Thm 0.1.9. Alternatively, since monotone function can have ^{only} countably simple discontinuity, if no such shared discontinuities with f and α , then countably discontinuous f is R.S. integrable. But, in this, prove by elementary, using properties of monotone.

Since α is continuous, for each $n \in \mathbb{N}$, there exists a partition $P_n \in \mathcal{P}[a, b]$,

$$\Delta \alpha_i = \frac{\alpha(b) - \alpha(a)}{n} \quad \text{for all } i$$

(More specifically, let $n \in \mathbb{N}$ be fixed. By the Intermediate Value Theorem, there exists $x_1 \in [a, b]$ s.t. $\alpha(x_1) = \alpha(a) + \frac{\alpha(b) - \alpha(a)}{n}$. Inductively,

there exists $x_2 \in [x_1, b]$ s.t. $\alpha(x_2) = \alpha(a) + 2 \cdot \frac{\alpha(b) - \alpha(a)}{n}$

$$\vdots$$

$$x_{n-1} \in [x_{n-2}, b] \text{ s.t. } \alpha(x_{n-1}) = \alpha(a) + (n-1) \cdot \frac{\alpha(b) - \alpha(a)}{n}$$

And take $x_n = b$. Essentially, the Existence such partition is obtained by I.V.P.

But, Monotone can have only first-kind discontinuity and

IVP implies it can not have first-kind discontinuity, thus Monotone I.V.T $\Rightarrow$ Continuity.

Given $\epsilon > 0$, ^{wlog,} Assume f is monotone increasing. Take $n \in \mathbb{N}$ s.t. $\frac{[\alpha(b) - \alpha(a)] \cdot [f(b) - f(a)]}{n} < \epsilon$.

$$\begin{aligned} U(f, P_n, \alpha) - L(f, P_n, \alpha) &= \sum_{i=1}^n [M_i - m_i] \cdot \Delta \alpha_i = \sum_{i=1}^n [f(x_i) - f(x_{i-1})] \cdot \Delta \alpha_i = \Delta \alpha \cdot \sum_{i=1}^n [f(x_i) - f(x_{i-1})] \\ &= \Delta \alpha \cdot [f(b) - f(a)] < \epsilon. \quad \square \end{aligned}$$

Proof of Thm. 0.1.10.

Since ϕ is continuous on the compact set $[m, M]$, it is continuous uniformly.

Given $\epsilon > 0$, there exists $0 < \delta < \epsilon$ s.t. $|s - t| < \delta \Rightarrow |\phi(s) - \phi(t)| < \epsilon$.

Since $f \in \mathcal{R}(\alpha)$ on $[a, b]$, there is a partition $\{x_0, \dots, x_n\}$ on $[a, b]$ s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \delta^2$$

Let $M_i = \sup f(t)$ and $m_i = \inf f(t)$ and $M_i^* = \sup(\phi \circ f(t))$ and $m_i^* = \inf(\phi \circ f(t))$ where the sup, inf values are taken in $t \in [x_{i-1}, x_i]$,

Divides the index set $\{1, \dots, n\}$ as:

$$A = \{i \in \{1, \dots, n\} \mid M_i - m_i < \delta\} \text{ and } B = \{i \in \{1, \dots, n\} \mid M_i - m_i \geq \delta\}.$$

Now, if $i \in A$, then $M_i^* - m_i^* = \sup(\phi \circ f(t)) - \inf(\phi \circ f(t)) \leq \epsilon$

because $\forall s, t \in [x_{i-1}, x_i]$, $|f(s) - f(t)| \leq M_i - m_i \Rightarrow |\phi(f(s)) - \phi(f(t))| < \epsilon$.

And, if $i \in B$, then $M_i^* - m_i^* \leq 2 \cdot \sup_{t \in [a, b]} |\phi(t)|$

We obtain that

$$\delta \cdot \sum_{i \in B} \Delta \alpha_i \leq \sum_{i \in B} [M_i - m_i] \cdot \Delta \alpha_i \leq \sum_{i=1}^n [M_i - m_i] \cdot \Delta \alpha_i = U(f, P, \alpha) - L(f, P, \alpha) < \delta^2.$$

That is $\sum_{i \in B} \Delta \alpha_i < \delta$. Finally,

$$\begin{aligned} \sum_{i=1}^n [M_i^* - m_i^*] \Delta \alpha_i &= \sum_{i \in A} [M_i^* - m_i^*] \Delta \alpha_i + \sum_{i \in B} [M_i^* - m_i^*] \Delta \alpha_i \\ &\leq \epsilon \cdot \sum_{i \in A} \Delta \alpha_i + 2 \cdot \sup_{t \in [m, M]} |\phi| \cdot \sum_{i \in B} \Delta \alpha_i \\ &\leq \epsilon \cdot [\alpha(b) - \alpha(a)] + 2 \sup |\phi| \cdot \delta \\ &\leq \epsilon \cdot [\alpha(b) - \alpha(a) + 2 \sup |\phi|] \end{aligned}$$

Proof of Thm 0.11.

Suppose that $f \in \mathcal{R}(a)$. Given $\epsilon > 0$, there is a partition $P = \{x_0, \dots, x_n\}$ on $[a, b]$ s.t.
 $U(f, P) - L(f, P) < \epsilon$, since f' is continuous, thus $f' \in \mathcal{R}$.

By the Mean-Value Theorem, for each $1 \leq i \leq n$

$$\Delta d_i = d(x_i) - d(x_{i-1}) = d'(t_i)(x_i - x_{i-1}) = d'(t_i) \cdot \Delta x_i \text{ for some } t_i \in [x_{i-1}, x_i]$$

For any $s_i \in [x_{i-1}, x_i]$,

$$\sum_{i=1}^n |d'(s_i) - d'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n [M_i - m_i] \cdot \Delta x_i < \epsilon.$$

$$\begin{aligned} \text{Now, } & \left| \sum_{i=1}^n f(s_i) \cdot \Delta d_i - \sum_{i=1}^n f(s_i) \cdot d'(s_i) \cdot \Delta x_i \right| \\ &= \left| \sum_{i=1}^n f(s_i) \cdot d'(t_i) \cdot \Delta x_i - \sum_{i=1}^n f(s_i) \cdot d'(s_i) \cdot \Delta x_i \right| \\ &= \left| \sum_{i=1}^n [d'(t_i) - d'(s_i)] \cdot f(s_i) \cdot \Delta x_i \right| \leq \sup |f| \cdot \epsilon \quad (*) \end{aligned}$$

$$\text{Therefore, } \left| \int_a^b f \, d\alpha - \int_a^b f d' \, dx \right| \leq |U(f, P, \alpha) - U(f d', P, x)| \leq \sup |f| \cdot \epsilon$$

Since (*) holds for any refinement of P .

0.1.3 Riemann-Integral

Definition 0.1.12. Let $\mathcal{P}[a, b]$ be a set of partitions. Define a Norm as:

$$\|\cdot\| : \mathcal{P}[a, b] \rightarrow \mathbb{R} : P \mapsto \max\{|x_i - x_{i-1}| \mid i = 1, 2, \dots, n\} \text{ where } P = \{x_0, x_1, \dots, x_n\}$$

Theorem 0.1.13. Let $f : [a, b] \rightarrow \mathbb{R}$ and $A \in \mathbb{R}$, TFAE:

1. f is Riemann-Integral with $\int_a^b f = A$.
2. For any $\varepsilon > 0$, there exists $\delta > 0$ such that $\forall P \in \mathcal{P}[a, b], \|P\| < \delta \implies |R(f, P) - A| < \varepsilon$.
3. For any $\varepsilon > 0$, there exists $\delta > 0$ such that $\forall P \in \mathcal{P}[a, b], P \supseteq P_0 \implies |R(f, P) - A| < \varepsilon$.

The statement 2) can be written as: $\lim_{\|P\| \rightarrow 0} R(f, P) = A$.

Proof of Thm 0.1.13.

Enough to show $1) \iff 2)$. Let $\int_a^b f = A$.

Given $\varepsilon > 0$, there exists $P_0 = \{x_0, x_1, \dots, x_n\}$ s.t. $U(f, P_0) < A + \varepsilon$.

Put $M = \sup |f|$, $\delta_1 = \frac{\varepsilon}{nM}$. Let $P \in \mathcal{P}[a, b]$ s.t. $\|P\| < \delta_1$. Write $P = \{y_0, \dots, y_m\}$.

Let $I = \{i \in \{1, \dots, m\} \mid (y_{i-1}, y_i) \cap P_0 \neq \emptyset\}$ ($M \geq 0, f > 0$. ($f = f + \min f$))

$$J = \{j \in \{1, \dots, m\} \mid (y_{j-1}, y_j) \cap P_0 = \emptyset\}$$

$$\text{Now, } U(f, P) = \sum_{i=1}^m M_i (y_i - y_{i-1})$$

$$= \sum_{i \in I} M_i (y_i - y_{i-1}) + \sum_{i \in J} M_i (y_i - y_{i-1})$$

$$< nM \|P\| + U(f, P_0) < A + 2\varepsilon.$$

점의 개수 많아서 n 개! 더 큰 구간에서 $\sup$!

Similarly $L(f, P) > A - 2\varepsilon$, for some $\delta_2 > 0$, $\|P\| < \delta_2$

Theorem 0.1.14. Let $f : [a, b] \rightarrow \mathbb{R}$ be a Riemann-Integrable function. Define

$$F : [a, b] \rightarrow \mathbb{R} : x \mapsto \int_a^x f(t)dt$$

Then, F is Continuous on $[a, b]$. Particular, if f is continuous at $x_0 \in [a, b]$, then F is differentiable at x_0 with

$$F'(x_0) = f(x_0)$$

Theorem 0.1.15. The fundamental theorem of calculus

If $f : [a, b] \rightarrow \mathbb{R}$ is Riemann-Integrable and there exists differentiable function $F : [a, b] \rightarrow \mathbb{R}$ such that $F' = f$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

0.1.4 Bounded Variation Function

Definition 0.1.16. Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function and $P = \{x_0, x_1, \dots, x_n\} \in \mathcal{P}[a, b]$. Define a Variation of f with respects to P as:

$$V_a^b(f, P) \stackrel{\text{def}}{=} \sum_{i=1}^n |f(x_i) - f(x_{i-1})|$$

If the set $\{V_a^b(f, P) \mid P \in \mathcal{P}[a, b]\}$ is bounded above, then f is called *Bounded Variation Function*. In this case, define a *Total Variation* of f as:

$$TV_a^b(f) \stackrel{\text{def}}{=} \sup\{V_a^b(f, P) \mid P \in \mathcal{P}[a, b]\}$$

In simply, denote $V(f, P) = V_a^b(f, P)$ and $TV(f) = TV_a^b(f)$.

Lemma 0.1.17. Let $f, g: [a, b] \rightarrow \mathbb{R}$ be Bounded Variation Functions, $\alpha \in \mathbb{R}$ be a real number. Then, $f + g, fg, \alpha f$ are Bounded Variation Functions and satisfy

$$\begin{aligned} TV(f + g) &\leq TV(f) + TV(g) \\ TV(fg) &\leq \sup_{t \in [a, b]} |g(t)| \cdot TV(f) + \sup_{t \in [a, b]} |f(t)| \cdot TV(g) \\ TV(\alpha f) &= |\alpha| \cdot TV(f) \end{aligned}$$

Lemma 0.1.18. Let $f: [a, b] \rightarrow \mathbb{R}$ be a map, and $a < c < b$.

f is bounded variation on $[a, b]$ if and only if f is bounded variation on $[a, c]$ and $[c, b]$

In this case, the following equality holds:

$$TV_a^c(f) + TV_c^b(f) = TV_a^b(f)$$

Theorem 0.1.19. Let $f: [a, b] \rightarrow \mathbb{R}$ be given.

f is bouned variation if and only if For some monotone increasing $g, h: [a, b] \rightarrow \mathbb{R}$, $f = g - h$

Proof. Suppose f is bounded variation. Define

$$g(x) = V_a^x(f) \quad \text{and} \quad h(x) = V_a^x(f) - f(x).$$

By definition, g is monotone increasing. And, if $a \leq x < y \leq b$, then

$$h(y) - h(x) = V_a^y(f) - V_a^x(f) - (f(y) - f(x)) = V_x^y(f) - (f(y) - f(x)) \geq 0.$$

The converse is trivial. $\square$

Proposition 0.1.20. Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuously differentiable function. If f is bounded variation, then

$$\int_a^b |f'(x)| dx \leq TV_a^b(f)$$

Proof. Let $P = \{x_0, \dots, x_n\}$ be a Partition on $[a, b]$. By the Mean-Value Theorem, for each $1 \leq i \leq n$, for some $t_i \in [x_{i-1}, x_i]$,

$$V_a^b(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i$$

Meanwhile,

$$L(|f'|, P) \leq \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \leq U(|f'|, P)$$

Both Lower Sum and Upper Sum converge to the $\int_a^b |f'(x)| dx$, we can complete this conversation.

Given $\varepsilon > 0$, there exists a partition P on $[a, b]$ such that

$$\int_a^b |f'(x)| dx \leq V_a^b(f, P) + \varepsilon \leq TV_a^b(f) + \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, therefore we obtain the result. □

Note: if f is continuously differentiable, then

For any $\varepsilon > 0$, there exists a partition P on $[a, b]$ such that $\left| \int_a^b |f'(x)| dx - V_a^b(f, P) \right| < \varepsilon$.

And, we can weaken the condition: continuously differentiable $\rightarrow$ differentiable and the derivative is Riemann-Integrable.

Proposition 0.1.21. If $f : [a, b] \rightarrow \mathbb{R}$ is differentiable and the derivative f' is bounded, then f is bounded variation.

Proof. For any partition P on $[a, b]$, by the Mean Value Theorem, there exists $t_i \in (x_{i-1}, x_i)$ such that

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n \sup |f'| \cdot \Delta x_i = \sup |f'| \cdot 1$$

Since P was arbitrary, f is bounded variation. □

Definition 0.1.22. Let $\alpha : [a, b] \rightarrow \mathbb{R}$ be a bounded variation.

A map f is said to be *Riemann-Stieltjes integrable* with respect to α if:

for some monotone increasing α_1, α_2 , $\alpha = \alpha_1 - \alpha_2$ such that $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$

and in this case, define

$$\int_a^b f d\alpha = \int_a^b f d\alpha_1 - \int_a^b f d\alpha_2$$

Lemma 0.1.23. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then $f \in \mathcal{R}(\alpha)$ for any bounded variation α .

Theorem 0.1.24. Let $f, \alpha : [a, b] \rightarrow \mathbb{R}$ be bounded variation functions. If $f \in \mathcal{R}(\alpha)$, then $\alpha \in \mathcal{R}(f)$ with

$$\int_a^b f d\alpha + \int_a^b \alpha df = f(b)\alpha(b) - f(a)\alpha(a)$$

Corollary 0.1.25. If $f : [a, b] \rightarrow \mathbb{R}$ is monotone increasing and $g : [a, b] \rightarrow \mathbb{R}$ is continuous. Then, there exists $c \in [a, b]$ such that

$$\int_a^b f(x)g(x)dx = f(a) \int_a^c g(x)dx + f(b) \int_c^b g(x)dx$$

0.1.5 Step function

Definition 0.1.26. Define a *Step function* as:

$$I: \mathbb{R} \rightarrow \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases}$$

Lemma 0.1.27. If $a < s < b$, f is bounded map on $[a, b]$, f is continuous at s , and $\alpha(x) = I(x - s)$, then

$$\int_a^b f d\alpha = f(s)$$

Theorem 0.1.28. Suppose that $c_n \geq 0$ and $\sum c_n$ converges. And, $\{s_n\}$ is a sequence of distinct points in (a, b) . Define

$$\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n)$$

If f is Continuous on $[a, b]$, then

$$\int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$$